A Supervised Progressive Growing Generative Adversarial Network for Remote Sensing Image Scene Classification

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***Abstract*— Remote sensing image scene classification is a challenging task. With the development of deep learning, methods based on convolutional neural networks (CNNs) have made great achievements in remote sensing image scene classification. Since the training of a CNN requires a large number of labeled samples, a generative adversarial network (GAN) for sample generation represents a new opportunity to solve the problem of the limited samples. However, most of the existing GAN-based sample generation methods can only generate unlabeled samples, instead of samples labeled with the corresponding scene category. In this article, to solve the problem, a supervised progressive growing generative adversarial network (SPG-GAN) is proposed for remote sensing image scene classification. The proposed method can generate labeled samples for the remote sensing image scene classification, significantly improving the classification accuracy in the case of limited samples. The SPG-GAN method has two main improvements. First, a conditional generative framework for labeled samples is proposed, in which the label information is added in the channel dimension as the input. By considering the constraints of the label information in the loss function, the network can be trained in the direction of a specific category. As a result, the network can generate remote sensing image scene classification samples with label categories. Second, a progressive growing sample generation method is introduced. In order to ensure that the generated samples have more spatial details, they are generated by progressively adding modules to the generator and discriminator, thereby ensuring that the generated sample is of better quality. After testing on two benchmark datasets and carrying out a large-scale experiment in the central area of the city of Wuhan in China, it was found that the proposed method can obtain a superior scene classification accuracy in the case of limited samples.**

***Index Terms*— Generative adversarial network (GAN), progressive growing, remote sensing image scene classification, sample generation.**

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I. Introduction

REMOTE sensing image scene classification is one of the most direct ways to understand remote sensing scenes. It involves labeling the given scene images with predefined semantic categories [1], [2], and has been widely used in urban planning [3], [4], geological disaster monitoring [5], [6], remote sensing image retrieval [7]–[9], and other fields [10], [11].

The early remote sensing image scene classification methods were mostly based on low-level visual features, such as the spectral, textural, and structural features. Consequently, remote sensing images are usually described by a feature vector extracted from the low-level visual attributes, either locally [12] or globally [13]. However, it is worth noting that the scene classification methods based on low-level visual features perform better on remote sensing scenes with a uniform structure and spatial arrangement, but they struggle to describe spatial distributions with high diversity and nonuniformity [14]. Subsequently, remote sensing image scene classification methods based on mid-level feature extraction have emerged. However, feature descriptors, such as the scale-invariant feature transform (SIFT) [15], texture descriptors [16], [17], and the color histogram, are manually designed descriptors. The probabilistic topic model is then used to encode the features obtained by these descriptors so that the local description features replace the entire scene image, in methods such as the vector of locally aggregated descriptors (VLADs) [18] and the bag-of-visual-words (BoVWs) model [19]–[21]. Finally, a classifier is used to complete the scene classification. However, these scene classification methods based on low- and mid-level features all require handcrafted features, and the representation capabilities of these handcrafted features are limited.

In recent years, deep learning has been widely used in various fields due to its ability to automatically extract feature information [22]–[24]. Deep learning has also been widely used in remote sensing image processing. The remote sensing image scene classification methods based on convolutional neural networks (CNNs) [25]–[27] and other more advanced deep learning technologies [28]–[30], in particular, have made remarkable achievements and have shown better performances

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than the traditional methods in the feature representation of remote sensing images. However, to achieve excellent performance, the methods based on deep learning require a large number of training samples, which results in a high cost for the sample labeling and the need for expert prior knowledge.

To summarize the above, it is the problem of training with limited samples. To date, there have been many methods proposed based on prior knowledge to solve the problem [31]. This kind of method mainly makes full use of the prior knowledge and experience to guide the learning progress of new tasks, which specifically includes metalearning [32], [33], metric learning [34], [35], and transfer learning [36], [37]. However, these methods rely on the feature representation ability of the model and require a source domain dataset or auxiliary dataset, which is complicated and unintuitive. Compared with these methods, a straightforward way to solve the limited samples learning problem is to increase the number of training samples by data augmentation. Data augmentation initially involved image transformation operations, such as simple geometric transformation, pixel change, or simple direct image fusion, to enhance the dataset. However, most of these methods require manual design and other operations [38], and they have a limited ability to increase the number and diversity of the samples. As another method of data augmentation, sample generation can solve the above problems. Sample generation can not only expand the number of samples but it can also enrich the content of the samples. These generated samples can improve the generalization ability of the model and suppress the risk of overfitting.

Generative adversarial networks (GANs) [39]–[41] are an emerging type of generative model, which can automatically learn the distribution of the original data by having the generator and discriminator compete with each other to generate fake data that are similar to the original data. As a result, GANs are an alternative way to generate remote sensing image scene classification samples to solve the sample generation problem for remote sensing image scene classification. For example, Lin *et al.* [42] applied a GAN to remote sensing image scene classification for the first time and proposed a multilayer feature matching generative adversarial network (MARTA-GAN), which can learn interpretable representations from remote sensing datasets to improve the classification accuracy; Pan *et al.* [43] proposed the diversity-GAN method that uses stepwise generation to obtain high-quality scene classification samples; and Yu *et al.* [44] designed the attention-GAN method to obtain a better scene classification performance by enhancing the representation capability of the discriminator. However, the above methods are unsupervised methods, which can only generate unlabeled samples to enhance the feature representation capability of the model. Because there is no category information for the generated samples, they cannot be used to represent the key features for the specific category. To address this issue, Ma *et al.* [45] designed the SiftingGAN method to generate a large number of labeled samples for scene classification, where the samples are passed through the filter of the generator and discriminator to ensure that the output samples have higher authenticity. However, due to the operators, such as the convolution and pooling operators, the network can only obtain global high-level semantic information, which leads to difficulty in the acquisition of spatial details. For scene classification samples, in addition to the global semantic information, the key features that can clearly distinguish the samples are the local spatial details.

In this article, to solve the above problems, a supervised progressive growing generative adversarial network (SPG-GAN) for remote sensing image scene classification is proposed. SPG-GAN can generate remote sensing image scene classification samples containing label information and can ensure that these samples have richer spatial structure and detailed texture information through hierarchical information extraction [46], [47]. The main contributions of our study are summarized as follows.

1) *Conditional Labeled Sample Generation Framework:* A conditional generative framework for labeled samples is proposed. Label information is added in the channel dimension as the input of the generator and discriminator network, and guides the model to generate remote sensing image samples containing label information. By considering the constraints of the label information in the loss function, the discriminator not only judges the authenticity of the generated samples but also judges whether the generated samples match the corresponding labels.
2) *Progressive Growing Sample Generation:* In order to ensure that the generated samples have more spatial details, a progressive growing sample generation strategy is designed, in which convolution and sampling modules are stepwise added to the generator and discriminator. As a result, the network can obtain the hierarchical scene features in different scales, ensuring that the single fake samples possess rich spatial structure information and detailed texture information, which improves the quality of the fake samples.
3) We evaluated the effectiveness of the proposed method on two widely used benchmarks—the UC Merced dataset and the aerial image dataset (AID) dataset—and conducted a large-scale experiment in the central area of the city of Wuhan in China. The experimental results show that the proposed method can obtain superior scene classification accuracy in the case of limited samples.

The rest of this article is arranged as follows. In Section II, the related works are briefly introduced. Section III describes the proposed method in detail. Section IV presents the comprehensive results obtained on the two benchmark datasets and the central area of the city of Wuhan. Finally, our conclusions are summarized in Section V.

II. Related Works

A. Generative Adversarial Network

The GAN was first proposed by Goodfellow *et al.* [39], and GANs have since been widely applied in image generation and other fields. The network structure of a GAN consists of a generator (G) and a discriminator (D). The purpose of the generator is to learn the real data distribution as much as

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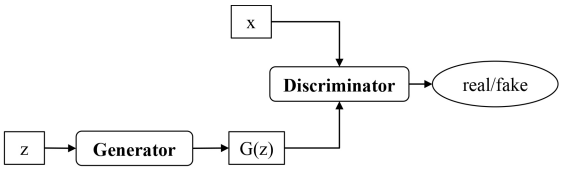

Fig. 1. GAN framework.

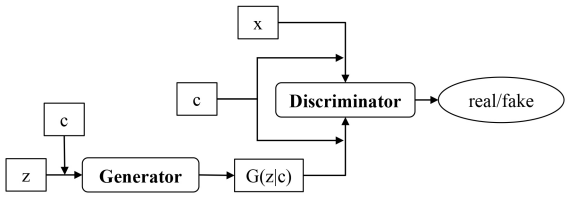

Fig. 2. CGAN framework.

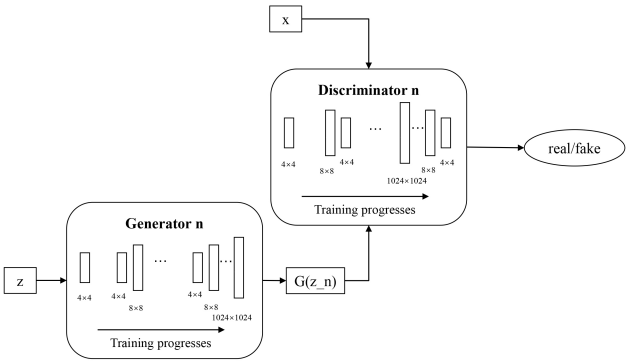

Fig. 3. PG-GAN framework.

possible, and the purpose of the discriminator is to correctly determine whether the input data come from the real data or the generator. In order to win this competition, the two participating networks need to be continuously optimized to improve their own generative and discriminative capabilities, which involves finding the Nash equilibrium between the two. A diagram of the GAN framework is shown in Fig. 1. In this process, the generator receives random noise z and generates fake data $G(z)$. The input of the discriminator is the real data x and fake data $G(z)$, and its output represents the probability of the discriminator believing that the input comes from real data. If the input comes from real data, the discriminator outputs 1; otherwise, it outputs 0. At the same time, the output of the discriminator is fed back to the generator to guide the training of the generator. The objective formula for the GAN is

$$\min_G \max_D V(D,G) = \mathrm{E}_{x\sim p_{data}(x)}\left[\log D(x)\right] + \mathrm{E}_{z\sim p_z(z)}\left[\log(1-D(G(z)))\right] \quad (1)$$

where E is the expectation operator, $p_{\mathrm{data}}(x)$ is the distribution of the real data, and $p_z(z)$ is a prior of the input noise z. This optimization process, which is similar to an adversarial game, ensures that the performance of the generator and discriminator continues to improve, and the final generator can learn the distribution of the real data, thereby generating high-quality fake samples.

However, the traditional GAN model is a typical unsupervised generative framework. The generated fake samples are also often random, without obvious label information.

B. Conditional Generative Adversarial Network

The traditional GAN model has almost no constraints on the generator, and the generated image categories are uncontrollable. To solve the problem, Mirza and Osindero [48] proposed the conditional generative adversarial network (CGAN) framework. The CGAN framework adds constraints to the structure of the traditional GAN framework, which ensures that the network training generates samples in a predetermined direction. A diagram of the CGAN framework is shown in Fig. 2. The difference with the traditional GAN model is that both the generator and the discriminator require input label information c corresponding to the training data during the training process. The label information c can be one-hot labels or image labels, which only needs to be merged with the input image when it is input into the network. Therefore, the GAN model at this time becomes a supervised conditional generative model. The generator must generate samples $G(z|c)$ that match the label c, and the discriminator not only distinguishes whether the image comes from a real sample or a fake sample, but also judges whether the image matches the label. The objective formula of the CGAN is

$$\min_G \max_D V(D,G) = \mathrm{E}_{x\sim p_{\mathrm{data}}(x)}\left[\log D(x|c)\right] + \mathrm{E}_{z\sim p_z(z)}\left[\log(1-D(G(z|c)))\right] \quad (2)$$

where c is the label. When compared with (1), conditional probability is used instead of unconditional probability, which can guarantee that samples of the corresponding category are generated.

Due to the traditional GAN being unsupervised, we cannot obtain the corresponding label information when generating remote sensing image scene classification samples. Therefore, the CGAN provides an opportunity, as it has the ability to generate samples with label information.

C. Progressive Growing Generative Adversarial Network

Due to the difficulty of high-resolution generation, the generated image can be easy to distinguish from a real image, which magnifies the gradient problem in GAN training. To address this issue, Karras *et al.* [46] proposed a new GAN—the progressive growing generative adversarial network (PG-GAN), the main idea of which is to gradually increase the scale of the generators and discriminators. First, the network starts training at a very low resolution and gradually adds new layers so that the model can improve the details during the training process. This approach not only speeds up the training process but also makes the training more stable and can generate extremely high-quality high-resolution images. As shown in Fig. 3, the progressive growing sample generation strategy enables the training to learn some large-scale structures of the image distribution in the initial stage and then shifts attention to finer and finer details, instead of learning all the scales at the same time, as in the traditional approach.

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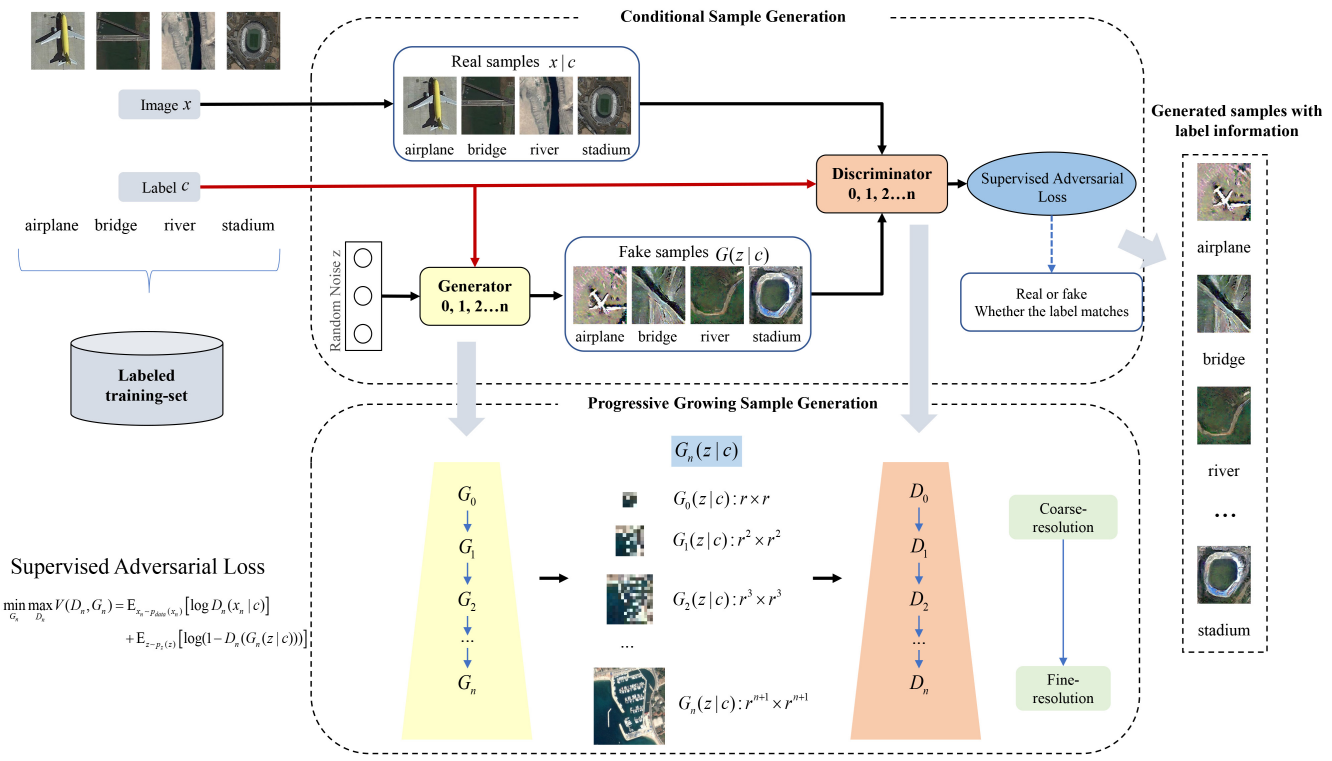

Fig. 4. Framework of the SPG-GAN.

The PG-GAN uses a network of generators and discriminators that are mirror images of each other and always grow synchronously. During the training process, all the layers of the two networks remain trainable. In addition, all the new layers are added smoothly, thus avoiding any sudden impact on the lower resolution layers that have already been trained.

However, the innovation of PG-GAN only proposes a progressive training approach. To perform conditional generation, AC-GAN [49], which is used to guide PG-GAN, is affected by the number of sample categories and the total number of samples, and the auxiliary classifier in AC-GAN biases the generator to generate easier-to-classify images [50]–[52], so it is not suitable for the generation of remote sensing image scene classification samples.

III. Proposed Method

A. Overview

In order to generate remote sensing image scene classification samples containing label information, we propose the SPG-GAN method, which has two main advantages compared with the other sample generation methods. First, it is a conditional sample generation framework. The framework is guided to generate remote sensing image samples containing label information by adding the label information to the input of the generator and discriminator network. At the same time, by considering the constraints of the label information in the loss function, the discriminator not only judges the authenticity of the generated samples but also judges whether the generated samples match the corresponding labels. Second, it is a progressive growing strategy of generating samples. Convolution and sampling modules are stepwise added to the generator and discriminator in the iterative process. This allows the samples to be gradually generated from a coarse resolution to a fine resolution, which ensures that the generated samples have rich spatial structure information and detailed feature information.

Fig. 4 illustrates the framework of the proposed method, in which the conditional sample generation and progressive growing sample generation modules are shown in detail. In the SPG-GAN training part, a small number of labeled remote sensing image scene classification samples are used as the original data for the training. The remote sensing images of the training set are combined with the corresponding labels as the input of the network, and the fake labeled samples are generated step by step from the coarse resolution to the fine resolution. Finally, after the SPG-GAN model has been trained, these generated labeled samples are combined with the original labeled samples, which can effectively increase the size of the dataset for training the classifier, so that a scene classification model can be trained. The SPG-GAN training strategy is formally described in Algorithm 1.

B. Conditional Sample Generation

The traditional GAN is a typical unsupervised generative model, which only needs unlabeled and out-of-order training data x as the input, so that it can learn the feature information of these data. After the adversarial training, the generator of the GAN can generate fake data $G(z)$ that are similar to the training data x. The generator of the traditional GAN aims to learn a mapping $f_G : z \rightarrow G(z)$. The generated data are, thus, random and do not contain any label information. As a result,

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Algorithm 1 SPG-GAN Train

Input: x : real image
c : one-hot vector corresponding to the label
z : random noise
Output: x' : fake image corresponding to the label

```
resolutions = [r_0, r_1, r_2, ....., r_N]
count = [e_0, e_1, e_2, ....., e_N]
for all i = 0 to N do
  for all j = 0 to count[i] do
    x' = G_i(z + c)
    lossD_i = BCE (D_i(x' + c) + D_i(x + c))
    Update D_i
    x' = G_i(z + c)
    lossG_i = BCE (D_i(x' + c))
    Update G_i
  end for
  G_{i+1} ← G_i
  D_{i+1} ← D_i
end for
```

they are not suitable to use as samples for remote sensing image scene classification.

The idea of the CGAN is introduced into the proposed framework, and the unsupervised generative model is transformed into a supervised conditional generative model. The conditioning is performed by feeding the conditional information c into both the discriminator and generator as an additional constraint. In a nutshell, what we need to do is to combine the remote sensing image scene label information c and the corresponding training image x in the channel dimension as the input of the network. In the conditional generative model, the prior input noise z and the condition information c jointly form a joint hidden layer representation. The adversarial training framework is quite flexible in terms of a hidden layer representation. Therefore, adding label information c is actually adding a potential constraint range representing label information to the random distribution of the GAN.

The conditional sample generation framework is shown in the upper part of Fig. 4. In iterative optimization, for the generator, the high-dimensional noise z and one-hot label c are merged in the channel dimension and input into the generator to obtain the fake samples $G(z|c)$ containing the category information. At this point, the aim of the generator is to learn the mapping

$$f_G : z + c \rightarrow G(z|c). \tag{3}$$

For the discriminator, it is also necessary to merge the input image x and its corresponding one-hot label c in the channel dimension and input this into the network. In this way, the discriminator not only needs to judge whether the generated sample $G(z|c)$ is authentic but it also needs to judge whether the category of the generated sample is consistent with the original label.

With the iterative optimization of the model, the label information of the original sample guides the direction of the model training, and the network learns the information of the corresponding category. In this way, when the training is completed, the required remote sensing image scene classification samples can be generated according to the labels set by the user.

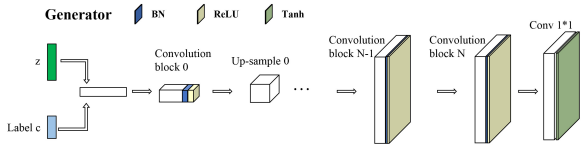

Fig. 5. Network structure of the generator.

C. Progressive Growing Sample Generation Strategy

In order to ensure that the generated remote sensing image scene classification samples have richer spatial structure information and detailed feature information, we adopt a progressive growing sample generation strategy, as shown in Fig. 4. The generator and discriminator carry out hierarchical information extraction, and scene samples are generated in a stepwise manner from a coarse resolution to a fine resolution.

1) Progressive Growing Sample Generator: The form of progressive growth is achieved by adding modules to the generator. As shown in Fig. 5, in the beginning, random noise z is merged with the label c in the channel. A specific convolution block C_0 with obvious padding is then used to process the merged tensor, and the result m_0 is the feature map at the zeroth level resolution with a resolution of $r \times r$. Therefore, the function of the first convolution module of the generator is

$$m_0 = C_0(z + c) \tag{4}$$

where C_0 is composed of a convolutional layer with a convolution kernel size of 4 and a padding size of 4, as well as a batch normalization layer and a rectified linear unit (ReLU) activation function. Although the feature map at this level of resolution does not have rich global structural information, it does contain the local texture information of the image, which directly determines the category of the remote sensing image scene classification samples.

When the first layer is trained for a certain number of times, the corresponding convolution block C_0 at the zeroth level of resolution can already fully learn the feature information at the current resolution. The feature map is upsampled to the resolution $r^2 \times r^2$, and the input of the next convolution block C_1 to be added is obtained. After being trained, the feature map under the current resolution already contains a large amount of spatial feature information. In this way, the relationship between the resolution size and the number of levels is

$$\text{size} = r^{n+1} \times r^{n+1} \quad (n = 0, 1, 2, \ldots, N). \tag{5}$$

After all the layers at different resolutions have been trained to a certain extent, we continue to add convolution blocks C_n to the generator and form a new generator G_n so that the learned detailed feature information and spatial structure information becomes more and more abundant. In this

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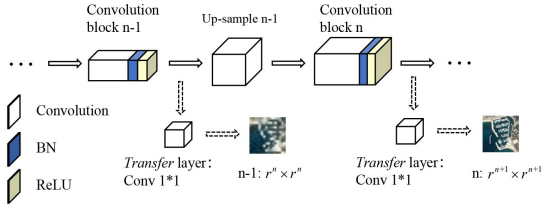

Fig. 6. Fake samples are output at different resolutions through the *Transfer* layer.

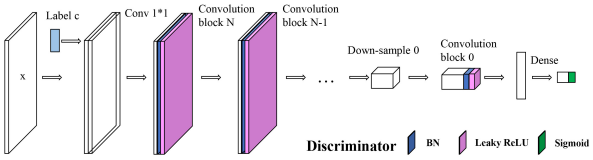

Fig. 7. Network structure of the discriminator.

way, upsampling blocks U_n and convolution blocks C_n are added iteratively until the size of the obtained feature map ($r^{n+1} \times r^{n+1}$) is consistent with the size of the original sample image. We then obtain the final feature map, which is the labeled fake sample image $G_N(z|c)$. The last layer of the generator is the Tanh activation function, and each of the convolution blocks uses a batch normalization layer and an ReLU activation function.

During the training process, the generated images $G_n(z|c)$ at different resolutions can be output through the *Transfer* layer (this is a convolutional layer with a kernel size of 1×1) from feature map m_n at any time, as shown in Fig.6. Of course, the function of the *Transfer* layer of the generator is

$$G_n(z|c) = \text{Transfer}(m_n) \tag{6}$$

where m_n is the multichannel feature map at different resolutions.

It is worth mentioning that, when the current convolution block is trained, the parameters of the corresponding convolution block at the previous resolution are fixed, so the computing power required to train the entire network is relatively small. In this way, fake samples at different resolutions are generated by adding modules in a stepwise manner, which can make the generated spatial structure information and detailed texture information more realistic and rich, so as to ensure the quality of the generated sample image.

2) Progressive Growing Discriminator: The discriminator also uses this stepwise strategy of adding modules for the training, as shown in Fig. 7. At each corresponding resolution, the original sample image x is first downsampled to the current resolution x_n, and then, the *Transfer* layer is used to increase the size of the channel dimension. In the subsequent operations, we use the same size convolution block as in the generator and the downsampling operation. The difference is that the discriminator uses the leaky ReLU activation function in its convolution block. The last layer of the discriminator is the *Dense* layer, which can obtain feature information of size 1. Finally, the sigmoid activation function is used to output the result. Regardless of whether it is a real sample or a fake sample, the corresponding one-hot labels are merged in the channel dimension when the data are input into the discriminator. At each resolution, the discriminator participates in the decision-making, which causes more feedback information to flow into the generator, so that the generator can generate higher authenticity samples.

The progressive growing sample generation strategy is an approach that can also suppress the collapse of the GAN training model. The generator and discriminator networks in the proposed method are mirror images of each other and always grow synchronously. During the training process, all the modules of the two networks remain in a trainable state. When new modules are added to the network, they are smoothly added, so as to avoid any sudden impact on the lower resolution layer that has already been trained. These generated images at a low resolution are also very stable, so that the discriminator is less dependent on the samples, which can successfully suppress the problem of pattern collapse during the training process.

D. Supervised Adversarial Loss

Label information is added as the input to the generator, as the purpose of the generator is to generate a scene sample with category information. Therefore, in the proposed method, the loss function of the generator is

$$\ell_{\text{adv}}^{G_n} = -\text{E}_{x'_n} \log\left(D\left(x'_n|\text{c}\right)\right) \tag{7}$$

where $D(x'_n)$ denotes the logits output from the last layer of the discriminator; x'_n represents the fake sample $G_n(z|c)$ generated by the generator at the nth level of resolution ($n = 0, 1, 2, \ldots, N$); and c is the category label information of the remote sensing image scene classification samples.

When the real sample or generated sample is input into the discriminator for identification, the label information also needs to be merged as input. Therefore, the discriminator not only needs to judge the authenticity of the input sample but also needs to judge whether the label information matches. By considering the label information, the loss function of the discriminator is

$$\begin{aligned}\ell_{\text{adv}}^{D_n} = &-\text{E}_{x_n} \log(D(x_n|\text{c})) \\ &-\text{E}_{x'_n} \log\left(1 - D\left(x'_n|\text{c}\right)\right)\end{aligned} \tag{8}$$

where x_n and x'_n, respectively, represent the real samples and the fake samples generated by the generator at the nth level of resolution.

Therefore, the objective function of SPG-GAN is

$$\begin{aligned}\min_{G_n} \max_{D_n} V(D_n, G_n) = &\text{E}_{x_n \sim p_{\text{data}}(x_n)}\left[\log D_n(x_n|c)\right] \\ &+\text{E}_{z \sim p_z(z)}\left[\log(1 - D_n(G_n(z|c)))\right].\end{aligned} \tag{9}$$

E. Scene Classification

The trained SPG-GAN model can be used to generate a large number of remote sensing image scene classification samples with label information to form a fake sample dataset. By combining the fake sample dataset with the original dataset by category, we can obtain a larger remote sensing image scene classification dataset.

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Finally, we can use remote sensing image scene classification methods, such as CNNs, to train with the new dataset and obtain the final results.

IV. Experimental Results And Analysis

A. Dataset Descriptions

In the experiments, we used two benchmark datasets, i.e., the UC Merced dataset and the AID dataset. A Google Earth image of the city of Wuhan in China was also used to carry out a large-scale experiment.

1) *UC Merced Dataset [19]:* The UC Merced dataset was released in 2010 and contains 21 scene categories. Each category is composed of 100 land-use images. The dataset has a total of 2100 scene images with a pixel resolution of 0.3 m. These images were manually extracted from the United States Geological Survey National Map Urban Area Imagery collection, covering the 21 regions of the United States. The image size is 256×256 pixels.
2) *AID Dataset [53]:* The AID dataset is a relatively large-scale aerial scene classification dataset. The dataset was released by Wuhan University in 2017 and has 30 scene categories. Each scene class is composed of 220 to 420 images that are cropped into a size of 600×600 pixels. These images were downsampled to 256×256 pixels in the experiments conducted in this study. The dataset contains a total of 10 000 scene images. Differing from the UC Merced dataset, the AID dataset is multisource, and the aerial images were collected by different satellites. In addition, the dataset is also multiresolution, and the pixel resolution of each scene category varies greatly, ranging from 0.5 to 8 m.
3) *Wuhan City Dataset [4]:* The image of the central area of the city of Wuhan is a fusion image made up of Google Earth images. The fusion image coverage is $15\,424 \times 12\,834$ pixels, and the image acquisition dates were April 23, 2017, and May 28, 2017. The spatial resolution of the imagery is 0.5 m. We cut the image into a large number of 256×256 patches for the land-use classification with 241 patches used as the training data and 1988 patches used as the test data.

B. Experimental Setup

1) Optimization:

All the experiments were implemented in the PyTorch deep learning framework using two Tesla V100 graphics processing units (GPUs).

For the two benchmark datasets, in order to simulate the case of limited labeled samples, we used 20, 40, and 60 samples from each type of sample as the training data, and the rest were used as the test data. SiftingGAN was used to generate a specific category of remote sensing image scene samples in the comparison experiments. We added the generated samples to the original dataset according to a ratio of 1:1 to form a new training dataset in all the experiments. The classifier used by both SiftingGAN and SPG-GAN was a VGG16 network without pretraining. Therefore, VGG16 was trained with the original dataset and served as the baseline for the experiments.

The experimental settings for each method are given as follows. In the sample generation part, the training generator and discriminator in the proposed method both used the Adam optimizer. The learning rate for both the generator and discriminator was set to 0.0002, and the batch size was set to 64. We set r to 2, and the number of levels was $N = 7$. However, to avoid the resolution being too low, we set the first resolution to 4, corresponding to different resolutions of 4, 8, 16, 32, 64, 128, and 256. The corresponding training times under each resolution were 0.3k, 0.6k, 1.2k, 3k, 6k, 12k, and 50k iterations. In the scene classification part, we used the VGG16 network without pretraining as the classifier, the optimizer was stochastic gradient descent (SGD), and the batch size was set to 48. All the methods were trained for 1000 epochs, with the learning rate for the first 700 epochs being 0.001, and the learning rate for the last 300 epochs being 0.0001.

SiftingGAN and SPG-GAN are supervised generative methods for remote sensing image scene classification. We also used MARTA-GAN, which is a representative unsupervised generative method, as another comparison method. MARTA-GAN extracts deep features from the remote sensing images by adding multiple feature layers in the GAN and uses the support vector machine (SVM) as the classifier for the scene classification. For MARTA-GAN, we set the training epochs to 600 and the penalty factor C of SVM to 100.

2) Metrics: All the methods were evaluated from two aspects. The visual effect obtained by human observation of the generated samples is an intuitive and effective evaluation method. If the generated sample matches the corresponding label and contains typical elements of the scene category in its image, the generation effect can be considered to be good. In addition, whether the contours of the objects in the generated samples are clear or blurred can also be used as a judgment criterion.

Two quantitative indicators were selected to evaluate the samples generated by the proposed method. The Fréchet inception distance (FID) score [54] is a measure used to calculate the distance between the feature vectors of the generated image and the original image. In the experiments, it was used to measure the similarity of the two groups of images. Its calculation formula is

$$d^2(F, G) = |\mu_X - \mu_Y|^2 + \mathrm{tr}\left[\sum_X + \sum_Y - 2\left(\sum_X \sum_Y\right)^{\frac{1}{2}}\right] \tag{10}$$

where μ_X, μ_Y and $\sum_X$, $\sum_Y$ are the mean and covariance matrices of the two eigenspace models of F and G, respectively. A lower score indicates that the two sets of images are more similar and means that the resulting samples are of high quality. The second quantitative indicator is the test accuracy of the remote sensing image scene classification model. If the generated samples are added, the scene classification accuracy is significantly improved, which indirectly reflects that the generated samples have richer remote sensing features. Here, the

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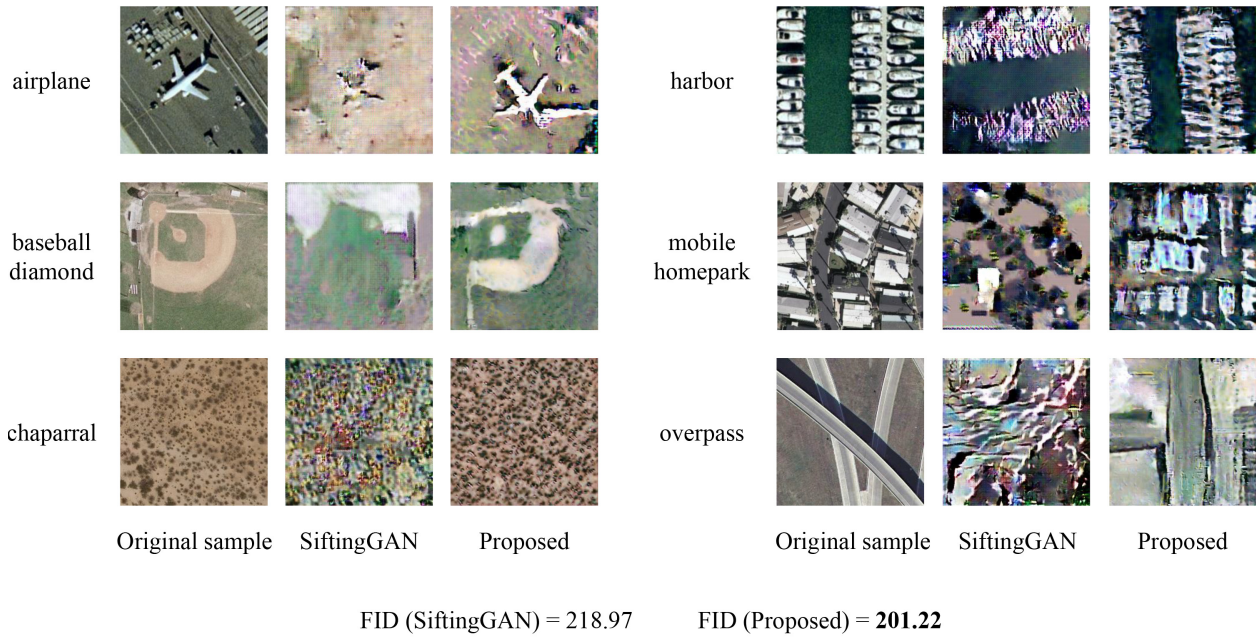

Fig. 8. Comparison between the generated samples and the real samples of the UC Merced dataset. We selected the six categories of *airplane*, *baseball diamond*, *chaparral*, *harbor*, *mobile homepark*, and *overpass* for display. From left to right are the real samples, the fake samples generated by SiftingGAN, and the fake samples generated by SPG-GAN.

overall accuracy (OA) and confusion matrices are specifically used as test accuracy indicators.

C. Experiment 1: UC Merced Dataset

1) Qualitative Evaluation: The authenticity of the generated remote sensing image scene classification samples directly determines the final classification accuracy, so we first compared the generated fake samples according to the category. Fig. 8 shows the real samples, the fake samples generated by SiftingGAN, and the fake samples generated by SPG-GAN for the UC Merced dataset. Since MARTA-GAN is an unsupervised generative method, it is not included in the visual comparison.

From the visual effect, it can be found that the authenticity of the samples generated by the proposed method is superior. For the sample categories with a simple image structure, such as *chaparral* and *harbor*, SiftingGAN can generate the general structure information of the sample. However, the proposed method can generate a scene sample with a close similarity to the original sample. The scene sample will also contain the complete global structure information and rich texture information. For the sample categories with complex image structures, such as *mobile homepark* and *overpass*, SiftingGAN cannot successfully generate these samples, whereas SPG-GAN can obtain a better visual effect, which can be easily resolved. Generating scene samples with complete global image structure and rich detail features is the key to improving the scene classification accuracy of remote sensing images. The overall structure and spatial texture details of the remote sensing image scene classification samples generated by the proposed method are more abundant, as it benefits from the progressive growing sample generation from different resolutions. The reason for this is that a deeper convolutional layer in the traditional CNN is more inclined to mine deeper information, such as global feature information. This is the case with the network structure of SiftingGAN, which makes it difficult to achieve sufficient structural diversity in the generated sample images. Therefore, with the parameters generated by the fake samples at different resolutions, the proposed method can ensure the overall structure of the samples and the retention of detailed feature information.

TABLE I
CLASSIFICATION RESULTS FOR THE UC MERCED DATASET

Method	Mode	OA(%)		
		20	40	60
Baseline	Supervised	62.98	73.09	78.09
MARTA-GAN	Unsupervised	42.97	54.36	61.84
SiftingGAN	Supervised	65.47	73.57	78.41
SPG-GAN	Supervised	**67.79**	**75.48**	**79.88**

In Fig. 8, we also show the FID scores of the samples generated by the two supervised methods, where it can be seen that the FID scores of the samples obtained using the proposed method are much lower than the FID scores of the samples obtained using SiftingGAN. This means that the authenticity of the samples generated by the proposed method is higher than that generated by SiftingGAN, which is consistent with the results of the visual interpretation.

2) Quantitative Evaluation: Table I lists the scene classification results obtained by the different methods for the UC Merced dataset. Because this experiment simulated remote sensing image scene classification in the case of limited labeled samples, the accuracy of the three methods is not very

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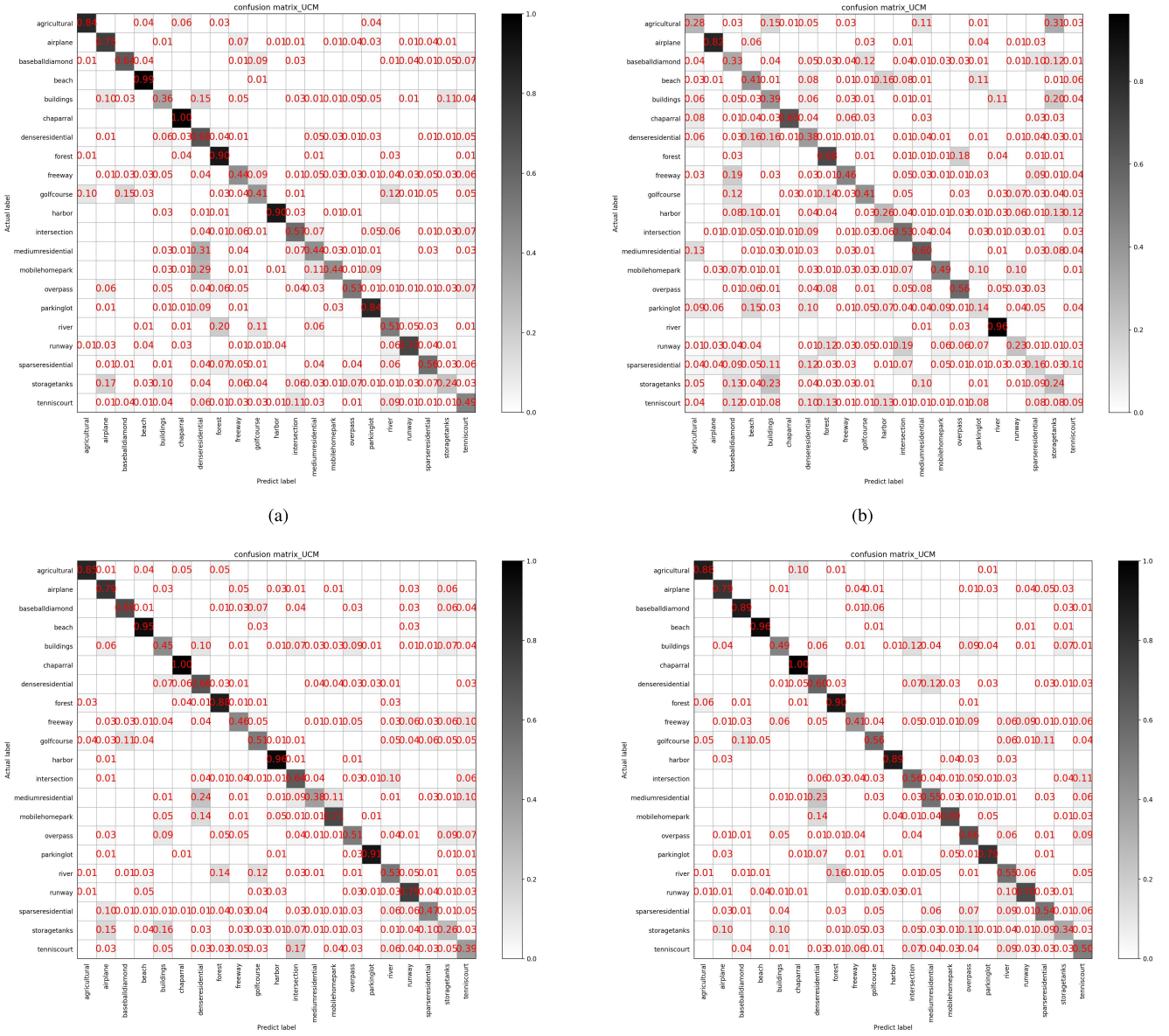

Fig. 9. Confusion matrix for the 20 sample training sets for the UC Merced dataset. From left to right are the results of (a) baseline method, (b) MARTA-GAN, (c) SiftingGAN, and (d) SPG-GAN.

high. In particular, the essence of MARTA-GAN is to obtain more high-level feature information by adding multiple feature layers to improve the accuracy of the remote sensing image scene classification. Moreover, the classifier of MARTA-GAN is the traditional SVM classifier, so its classification accuracy is very low. When 20 samples from each category of the UC Merced dataset participate in the training process, the classification accuracy of MARTA-GAN is only 42.97%. Therefore, in the case of remote sensing image scene classification with limited samples, the supervised generative approach is more effective than the unsupervised feature extraction approach.

On the other hand, due to the increase in the number of samples, SiftingGAN and the proposed method show an accuracy improvement of 2.49% and 4.81%, respectively, over the baseline method in the case of 20 samples per category. More importantly, the OA of the proposed method is improved by 2.32% compared with SiftingGAN. At the same time, it is worth noting that, when the number of original samples involved in the training increases, the increase in the accuracy of the results obtained by SiftingGAN is less than that of the baseline method. For the case of 60 training samples per category, the accuracy of SiftingGAN is only 0.32% higher than the baseline method, while the accuracy of the proposed method is still 1.79% higher, which is mainly because the structure and details of the samples generated by the progressive growing sample generation strategy in the proposed method are more complete.

The confusion matrix for scene classification with the UC Merced dataset, with 20 samples for each category as the training dataset, is shown in Fig. 9. For MARTA-GAN, the

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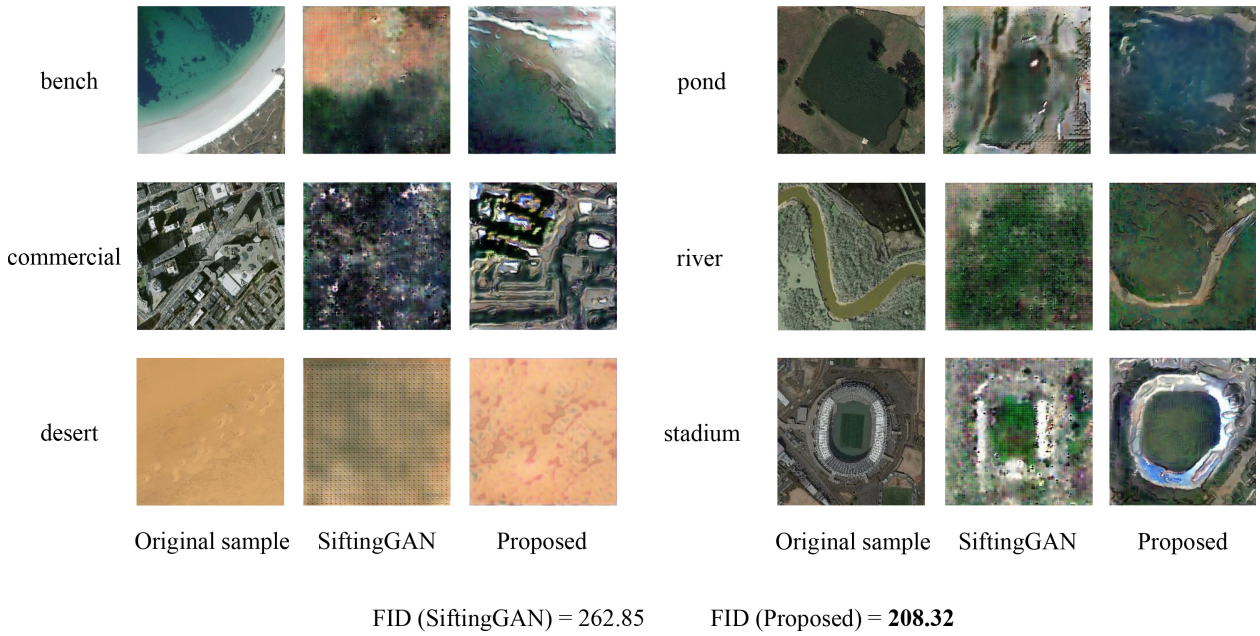

Fig. 10. Comparison between the generated samples and the real samples for the AID dataset. We selected the six categories of *bench*, *commercial*, *desert*, *pond*, *river*, and *stadium* for display. From left to right are the real samples, the fake samples generated by SiftingGAN, and the fake samples generated by SPG-GAN.

classification accuracy of almost every class is lower than that for the other three methods, which is consistent with the trend of the OA. For the supervised generative methods, we can find that some highly confusing categories are prone to misclassification, such as *baseball diamond* and *golf course*. In the confusion matrix obtained by the baseline method, a large part of the *golf course* category is incorrectly classified as *baseball diamond*, but the addition of the generated samples improves the classification accuracy between these two categories, to a certain extent. However, SPG-GAN shows a greater improvement than SiftingGAN, so SPG-GAN shows a good effect in dealing with similar features. At the same time, some of the categories have a more complex image structure, and the classification accuracy obtained by the baseline method is not very high. This is because the classifier needs to learn the more complex features in categories such as *buildings*, *medium residential*, and *mobile homepark*. By observing the classification accuracies of SiftingGAN and SPG-GAN in these categories, it can be found that the classification accuracy of SPG-GAN is greatly improved. This is because, for these categories, the generated samples should be more inclined to obtain the overall structure and detailed features of the image, so the SPG-GAN samples generated through multilevel resolution can effectively learn the information of both the low-level and deep features. This finding is consistent with our conclusion when we visually judged the authenticity of the generated samples.

D. Experiment 2: AID Dataset

1) Qualitative Evaluation: Fig. 10 shows the original samples of the AID dataset, the fake samples generated by SiftingGAN and the fake samples generated by the proposed method. For example, for the *bench* and *desert* categories, the samples generated by SiftingGAN all contain a small amount of noise. For these relatively simple remote sensing image scene samples with uncomplicated textures, the proposed method can generate fake samples that are very close to the original samples. For samples with clear objects, such as the *pond*, *river*, and *stadium* categories, SiftingGAN cannot obtain clear outlines of these objects. Although there is still a certain gap between the results generated by the proposed method and the original samples, the boundary contours of these objects can still be clearly observed. In addition, it can be observed that the FID score for the fake samples generated by SiftingGAN is as high as 262.85, whereas the FID score for the proposed method is only 208.32, which shows that the samples generated by the proposed method are more authentic. Therefore, for samples of both a simple scene and samples for scenes with a complex texture structure, the fake samples generated by the proposed method are better than those generated by SiftingGAN. This is because the proposed method adopts a progressive growing strategy, which considers feature information of different resolutions. This not only ensures the integrity of the global spatial structure but also ensures the richness of the detailed textures.

2) Quantitative Evaluation: Table II lists the scene classification results obtained by the different methods for the AID dataset. Among the results, the scene classification accuracy of MARTA-GAN is still the lowest at only 31.42%, 45.32%, and 50.67%, for 20, 40, and 60 samples of each category, respectively. As with the UC Merced dataset, the proposed method achieves the best results in the experiments with different sample sizes. Compared with the baseline method, the proposed method shows an improvement in accuracy of

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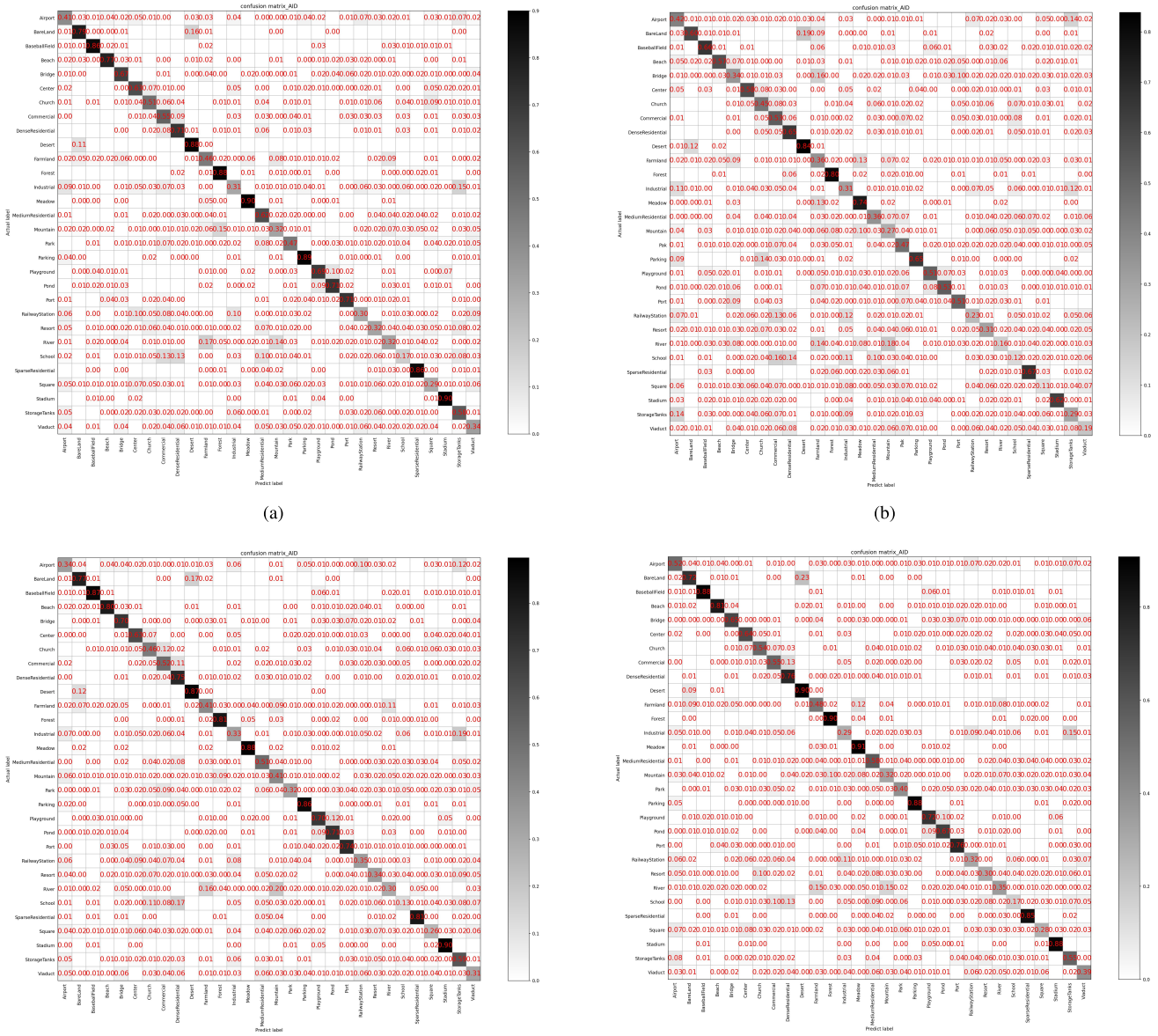

Fig. 11. Confusion matrix for the 40 sample training sets for the AID dataset. From left to right are the results of (a) baseline, (b) MARTA-GAN, (c) SiftingGAN, and (d) SPG-GAN.

TABLE II
CLASSIFICATION RESULTS FOR THE AID DATASET

Method	Mode	OA(%)		
		20	40	60
Baseline	Supervised	44.85	58.61	64.04
MARTA-GAN	Unsupervised	31.42	45.32	50.67
SiftingGAN	Supervised	45.51	59.12	64.27
SPG-GAN	Supervised	**47.19**	**59.34**	**65.13**

2.34%, 0.73%, and 1.09%, while SiftingGAN only shows an improvement of 0.66%, 0.51%, and 0.23%.

The confusion matrix for scene classification is shown in Fig. 11 for 40 samples for each category as the training dataset for the AID dataset. From the confusion matrix, we can see that the overall classification accuracy of SPG-GAN is higher than that of the baseline method, MARTA-GAN, and SiftingGAN. For the simple scenes, we take the *desert* category as an example. In this case, the classification accuracy of the baseline method is 88%, the classification accuracy of MARTA-GAN is 84%, the classification accuracy of SiftingGAN is 87%, and the classification accuracy of the proposed method is as high as 90%. This shows that the *desert* samples generated by the proposed method can effectively improve the classification accuracy for this category. However, the classification accuracy of SiftingGAN is lower than that of the baseline method, indicating that poor-quality fake samples can cause the original classification accuracy to decrease. For a complex residential area, we take the *sparse residential* category as an example. In this case, the classification accuracies of MARTA-GAN, SiftingGAN, and the proposed

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Fig. 12. Google Earth image and its ground truth for the central area of the city of Wuhan.

method are 67%, 81%, and 85%, respectively, which are all lower than the baseline method classification accuracy of 86%. For more complex scenes, it is difficult to learn the feature distribution, which makes the quality of the generated samples poor, but the proposed method still achieves the best results among the three methods. This is because the proposed method comprehensively considers the features at different resolutions. The effective combination of the global structural information and the local spatial texture can ensure that the generated samples are closer to the original samples.

E. Experiment 3: Cartographic Applications

In order to further verify the effectiveness of the proposed method, we undertook a large-scale experiment in land-use mapping for the central area of the city of Wuhan.

The image data used in the experiment and the ground truth for the land-use mapping are shown in Fig. 12. We cut the image into a large number of 256 × 256 patches for the scene classification. We then divided the 2229 patches into 24 scene categories for training and testing the model, 241 of which were used as the training dataset, and the rest were used as the test dataset. Since the number of patches per category obtained by cropping was different, we randomly selected 20% of the total number of samples per category as training samples, with each category not exceeding 20 samples. MARTA-GAN and SiftingGAN were again used as the comparison methods. After the training, we used the different methods to classify the scenes of all the patches and constructed a land-use map according to the classification results. According to [4], the land-use types were divided into 12 types, and their corresponding relationship with the scene categories is shown in Table III.

TABLE III
CATEGORY MAPPING DICTIONARY BETWEEN THE LAND-USE TYPES AND THE SCENE CATEGORY NAME

Code	Land-use type name	Scene category name
0	Water	*bridge*, *harbor*, *lake*, *ship*, *water*
1	Residential area	*dense-residential*, *medium-residential*, *sparse-residential*
2	Park	*park*, *meadow*
3	Transportation	*parking-lot*, *railway-station*
4	Education and science	*basketball-court*, *playground*, *school*, *tennis-court*
5	Farmland and forest	*farmland*
6	Green buffer	*greenbelt*
7	Cultural facilities	*center*
8	Square	*square*
9	Sports	*stadium*
10	Commercial and business facilities	*commercial*
11	Other non-construction land	*bare-land*
12	Industrial area	*industrial*

The scene classification results are listed in Table IV. Due to the large quantity difference between categories, we adopted the OA and kappa coefficient as the evaluation indices to evaluate the accuracy between categories. From the table,

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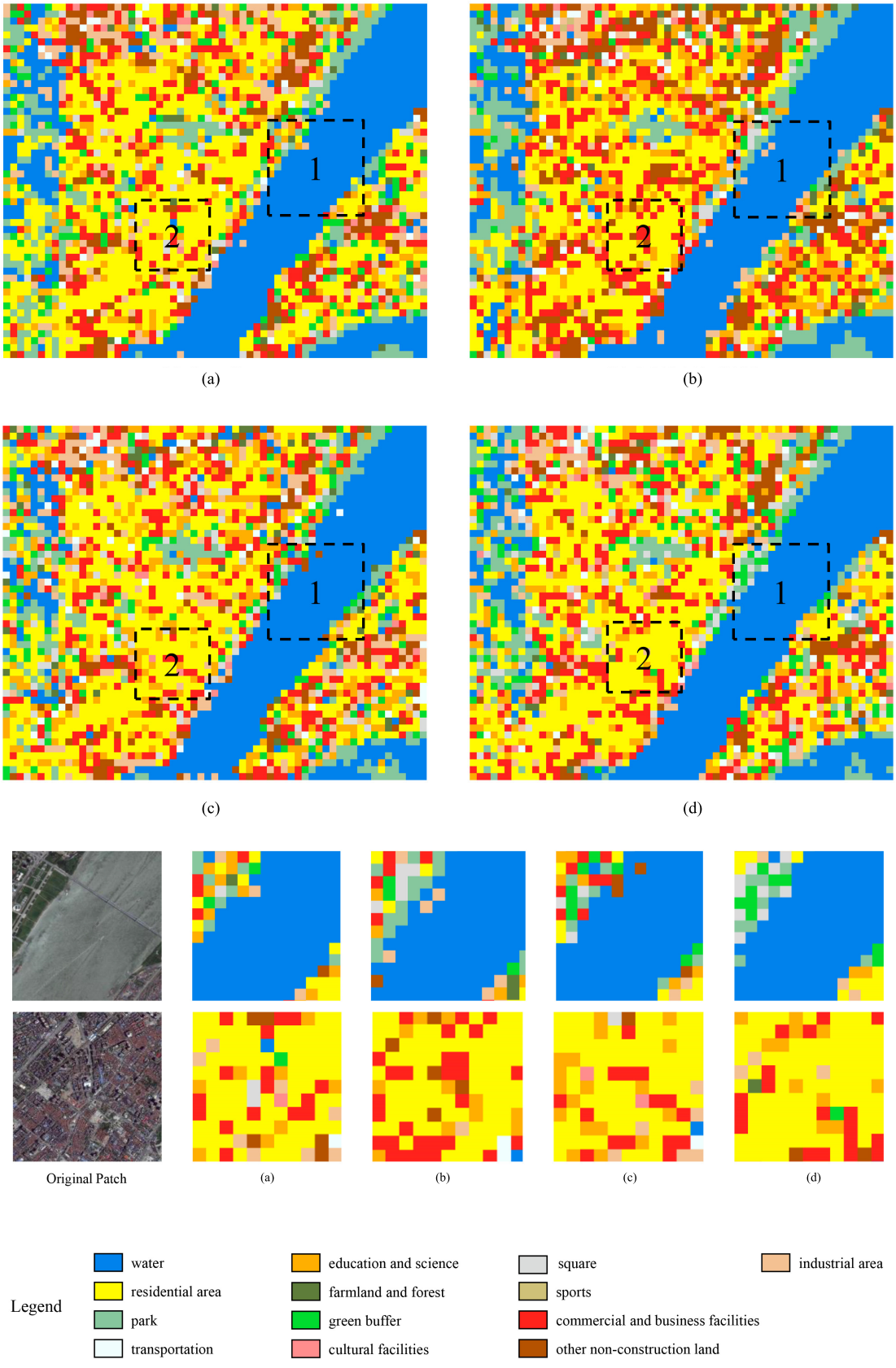

Fig. 13. Land-use maps of the central area of the city of Wuhan based on the different methods of scene classification. (a) Baseline. (b) MARTA-GAN. (c) SiftingGAN. (d) SPG-GAN.

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TABLE IV
SCENE CLASSIFICATION RESULTS FOR THE CENTRAL AREA OF THE CITY OF WUHAN

Method	Mode	OA(%)	Kappa
Baseline	Supervised	52.16	0.4472
MARTA-GAN	Unsupervised	54.07	0.4731
SiftingGAN	Supervised	57.64	0.5074
SPG-GAN	Supervised	**59.21**	**0.5278**

TABLE V
SIZE OF THE TRAINING PARAMETERS AND LENGTH OF THE INFERENCE TIME

Method	Mode	Parameters size(M)	Inference time(s)
MARTA-GAN	Unsupervised	87.3	2.842
SiftingGAN	Supervised	36.4	**1.218**
SPG-GAN	Supervised	**31.9**	2.080

it can be seen that the baseline method classification accuracy is the lowest, at 52.16%, and the classification accuracies of MARTA-GAN, SiftingGAN, and the proposed method are 54.07%, 57.64%, and 59.21%, respectively. In addition, the comparison of the kappa coefficients is consistent with the classification accuracy. From the accuracy analysis, the conclusion is consistent with the conclusion reached for the two benchmark datasets, in which SPG-GAN obtains the highest classification accuracy among all the methods. Similarly, both SiftingGAN and the proposed method show a certain degree of improvement in accuracy compared to the baseline method, which is due to the addition of the generated samples. However, since the samples generated by the proposed method have richer image structure information and detailed feature information, the accuracy is further improved by about 7.05%.

Fig. 13 shows the land-use mapping results of the four methods. It can be found that the mapping results are generally consistent, and the four methods all show a good mapping effect. However, through careful observation, it can be found that the proposed method obtains the best visual effect. In the area where the land-use type is *water*, the proposed method has no patch classification error, while the other methods all have patches of other land-use types in the water area, as shown in Zoom 1 in Fig. 13. When we observe the area near the river bank, the original image shows that most of the land-use types are the green land-use types, such as *square*, *park*, and *green buffer*. Among the land-use mapping results of the four methods, the proposed method is more accurate in this aspect, and other different land-use types appear near the river bank in the results of the three other methods. In addition, by observing the patch area shown in Zoom 2, it can be found from the original image that most of this area is a residential area, and the result obtained by the proposed method is mostly the *dense residential* category. The results of the other methods contain varying degrees of classification errors, as shown in the yellow area in the figure.

F. Discussion

1) Model Complexity and Inference Speed Analysis: Table V lists the sizes of the training parameters (only the parameters of the generator) of the different methods and the inference times. All the experiments were carried out under the same hardware and software environment. It can be seen that the training parameters of the proposed method are the smallest. Although the network structure of the proposed method is gradually added in the form of modules from coarse to fine, the training parameters of the final model are still very small, and no other redundant parameters are introduced.

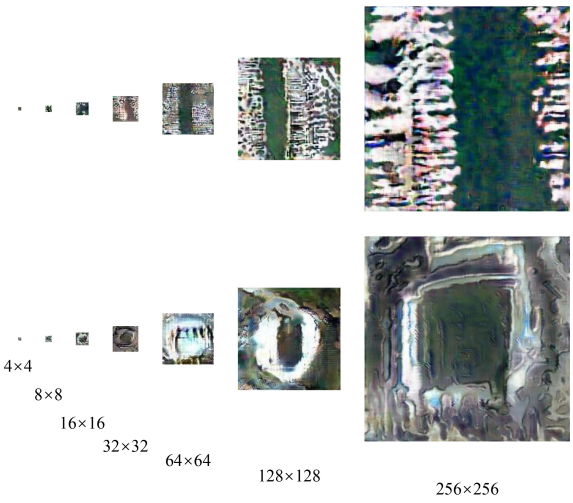

Fig. 14. Scenic classification samples of remote sensing images generated under different resolutions, taking *harbor* and *stadium* categories as examples.

From the loading of the model parameters to the final generation of the 8×8 sample map, the total time taken is the overall inference time. Since the proposed method needs to load parameters step by step, according to the resolution and each module, and the fact that it also includes some upper sampling layers, it takes longer to generate samples than SiftingGAN, but it takes less time than MARTA-GAN.

2) Generation of Samples at Different Resolutions: The progressive growing sample generation strategy is aimed at ensuring that the generated remote sensing image scene classification samples have richer spatial structure and detailed texture information. Fig. 14 shows the samples generated at different resolutions for the two categories of *harbor* and *stadium* in the two benchmark datasets. From the figure, it can be seen that the sample is generated step by step from a low resolution to a high resolution, which not only ensures that the semantic information of the generated sample is unchanged but also that the detailed texture information of the sample image becomes more and more abundant with the continuous improvement of the generation resolution. When generating low-resolution images, such as 16×16 and 32×32, the basic structure of the sample image has been shaped, and the semantic information of the generated image has been basically determined. After these slightly rough images are processed through high-resolution refinement, the

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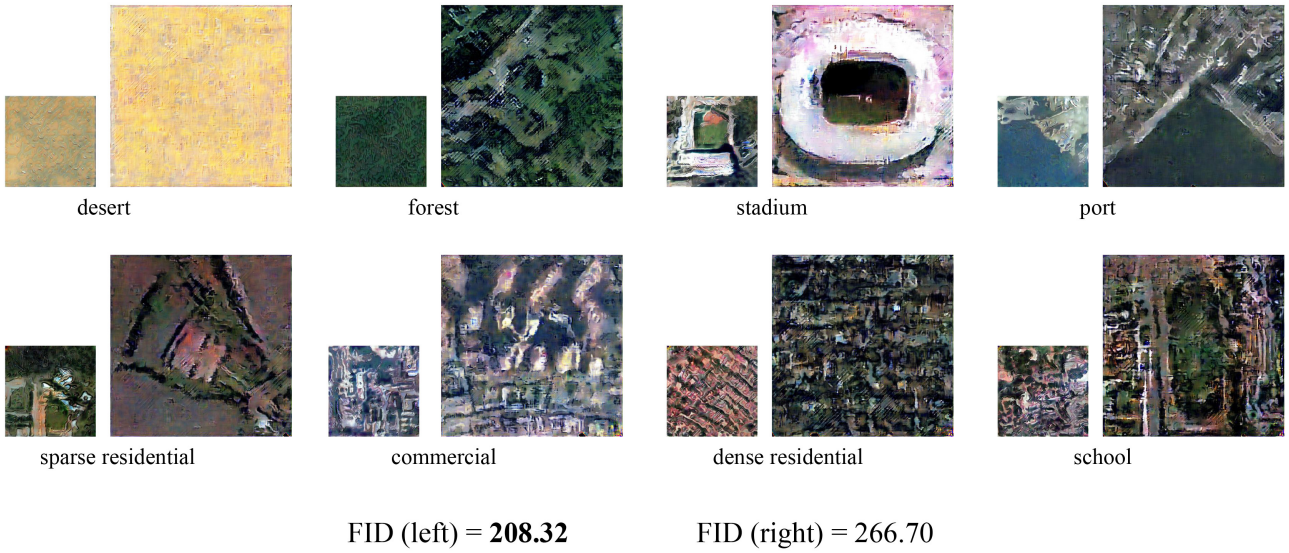

Fig. 15. Large-scale high-resolution image generation on the AID dataset, which is 512 × 512 in size. The size of the image on the left is 256 × 256.

texture structure of the ground objects in the image can be enriched, making the generated samples more realistic.

Since the proposed method adopts the progressive growing sample generation strategy, it is more effective for large-scale high-resolution image generation. However, due to the lack of real remote sensing image scene classification samples of a relatively large scale, we conducted a generation experiment on the original image size of the AID dataset. The result is shown in Fig. 15. The corresponding image size is 512 × 512. It can be seen that the generated samples have certain remote sensing scene category information and image texture detail information, and have benefited the multiresolution supervision and the step-by-step generation strategy from coarse to fine. At the same time, we output the FID scores of the generated images at two different scales. In general, the authenticity of the images generated under a high resolution is worse than that generated under a low resolution.

3) Random Noise and Category Conditions: Due to the addition of the CGAN, the conditional information imposes certain constraints on the results of the generated samples. Therefore, we conducted ablation experiments on the influence of the random noise z and category conditions c on the generated samples, as shown in Fig. 16.

The UC Merced dataset was used as the benchmark to test the generator output in the following two situations: lock c and change z, or lock z and change c. According to the experimental results, when the category condition information c is locked and the random noise z is changed, it means that samples of the same category are generated. In this way, the obtained sample image has very high homogeneity, and only some small details have different textures. In contrast, when z is locked and c is changed, it can be found that the generated samples still have rich category information. In particular, when observing a single generated sample, the image obtained by the proposed method is rich in spatial structure and texture information.

Therefore, we conclude that, in the conditional generative model, the multidimensional random noise z and the category condition c are combined in the channel dimension and sent to the generator to guide the conditional sample generation, which can indeed obtain samples with rich categories. However, the category condition plays a decisive role, which often leads to the problem of a lack of diversity among the generated samples of the same category. The elements of "condition" and "randomness" cannot be achieved at the same time because these two elements are inherently restrictive and contradictory. In order to solve this problem, we change the samples' output mode to produce more diverse samples. Instead of terminating the training process when the model is well trained, we adopt the method of outputting samples during training, continuously updating the network and generating samples synchronously. In this way, the model parameters of the generator network are always changing, which is equivalent to using a large number of models to generate samples, so as to ensure that the generated samples of the same category also have rich diversity.

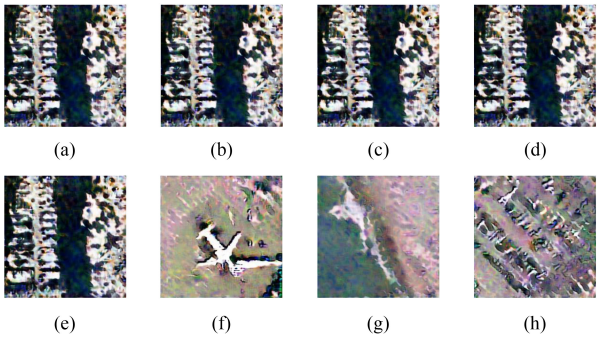

Fig. 16. Ablation experiment of random noise and category conditions. (a)–(d) Generated sample with locked c and changed z. (e)–(h) Generated sample with locked z and changed c.

4) Limitations of the Generated Samples: In the process of sample generation, there will of course be generation failure cases, which is due to the uncertainty of the random noise sampling and the complexity of the image texture in some categories. Fig. 17 shows the failed samples generated in

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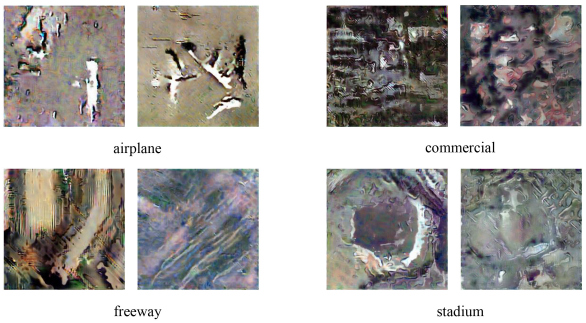

Fig. 17. Generated failed remote sensing image scene classification samples.

four categories: *airplane*, *commercial*, *freeway*, and *stadium*. In the *airplane* sample, a complete airplane is not generated, and there is no obvious corresponding building profile in the *commercial* sample. The roads in the *freeway* sample are mixed, they have no obvious boundaries, and there are breaks in the middle. In the *stadium* sample, the stadium and its surrounding elements are not well generated and displayed. Therefore, not all the generated samples are suitable for expanding the original sample set. Not only are such failed samples unable to improve the accuracy of remote sensing image scene classification in the case of limited samples but also cause more confusion and difficulty in distinguishing between categories.

V. Conclusion

In this article, we have proposed the SPG-GAN for remote sensing image scene classification. The proposed framework can generate labeled samples for remote sensing image scene classification, and the generated fake samples can then be combined with the original samples to form a larger dataset. This approach can significantly improve the remote sensing image scene classification accuracy in the case of limited labeled samples. The proposed method uses conditional sample generation to generate samples of the corresponding categories. Meanwhile, the strategy of progressive growing generates fake samples from the different resolutions by adding modules to the generator and discriminator in a stepwise manner. This effectively captures the structural and detailed information of the samples and ensures that the generated samples are of superior quality. However, there is still a gap between the samples generated by the proposed method and real samples. We will, therefore, continue to attempt to improve the network structure in the future to generate higher authenticity fake samples.

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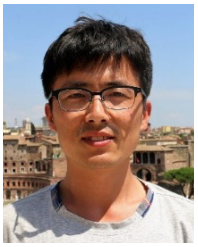

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